

Amazon Customer Behavior Analysis

Executive Summary

Project Overview

This project analyzes Amazon customer behavior to identify purchasing patterns, customer engagement trends, shopping satisfaction, and opportunities for improving customer experience. The analysis was performed using a cleaned survey dataset containing **800 customer records** across **24 business attributes**, covering customer demographics, purchasing behavior, browsing habits, product reviews, recommendation systems, and shopping satisfaction.

The project combined **data cleaning, exploratory data analysis (EDA), customer segmentation, statistical hypothesis testing, and business visualization** to transform raw customer responses into actionable business insights. The objective was to support data-driven decision-making by identifying customer segments, evaluating recommendation effectiveness, and proposing strategic business improvements.

Task 1: Data Cleaning & Preparation

Objective

Prepare the raw customer survey data for reliable analysis by improving data quality, standardizing variables, and engineering features required for downstream analytics.

Key Findings

- The original dataset contained **800 customer records** and **24 variables**, providing a comprehensive view of customer demographics and shopping behavior.
- **157 missing values** were identified in the **Product_Search_Method** column and were replaced with **"Not Specified"**, ensuring that all survey responses were preserved for analysis.
- **No fully duplicated records** were found in the dataset. Although duplicate transaction IDs were identified, they represented different customer responses and were therefore retained.

- Timestamp values were converted into a proper datetime format, column names were standardized, and new analytical features such as **Purchase_Categories_List** and **Personalized_Recommendation_Score** were created to improve analysis readiness.
- After preprocessing, the final dataset contained **800 complete records** with **25 analytical variables**, ready for exploratory analysis and modeling.

Business Impact

- Reliable data preparation improves the accuracy of business reporting and downstream analytical models.
- Preserving incomplete survey responses maximizes available customer information while minimizing unnecessary data loss.
- Standardized and well-structured data enables consistent reporting, dashboard creation, and future predictive analytics initiatives.

Task 2: Descriptive Statistics & Exploratory Data Analysis

Objective

Explore customer demographics and purchasing behavior to identify patterns in shopping preferences, browsing habits, and customer engagement.

Key Findings

- The dataset represents customers ranging from **3 to 67 years of age**, with an **average age of approximately 35 years**, indicating strong representation from young and middle-aged shoppers.
- Purchase behavior was distributed across multiple categories, with **Clothing and Fashion recording the highest purchase count (450)**, followed by **Others (412)**, **Home and Kitchen (391)**, **Beauty and Personal Care (383)**, and **Groceries and Gourmet Food (369)**.
- Customer purchasing frequency was relatively balanced, with **Once a Month** emerging as the most common purchase pattern, while a substantial proportion of customers also purchased weekly or multiple times per week.
- Gender representation was well balanced across all categories, reducing demographic bias within the analysis.

Business Impact

- Clothing and Fashion should remain a strategic product category due to its consistently high customer demand.
- Balanced purchasing behavior across multiple categories supports maintaining a diversified product portfolio rather than focusing on a single product segment.
- Customer segmentation strategies can be applied without significant demographic bias due to the balanced gender representation.

Task 3: Customer Segmentation & Profiling

Objective

Identify meaningful customer groups based on purchasing behavior and shopping satisfaction to support targeted marketing and retention strategies.

Key Findings

- Rule-based customer profiling identified two primary customer groups: **Regular Customers** and **At-Risk Customers**, with Regular Customers representing the larger portion of the dataset.
- K-Means clustering further identified **three distinct customer segments** based on purchase frequency and shopping satisfaction.
- **Cluster 2** represented the highest-value customer segment, recording the **highest average Purchase Score (4.59)** alongside a **Shopping Satisfaction score of 3.74**, indicating highly engaged and valuable customers.
- **Cluster 1** recorded the **lowest Shopping Satisfaction score (1.48)** despite maintaining a moderate **Purchase Score (2.66)**, highlighting customers who continue purchasing despite poor experiences and are therefore most susceptible to churn.
- **Cluster 0** recorded the **highest average Shopping Satisfaction (4.03)** while maintaining the **lowest Purchase Score (1.96)**, suggesting an opportunity to encourage more frequent purchases among otherwise satisfied customers.

Business Impact

- High-value customers should be prioritized through loyalty programs, exclusive offers, and personalized engagement.
- At-risk customers require proactive retention initiatives focusing on service quality, pricing, and customer support improvements.

- Satisfied but infrequent shoppers represent an opportunity for personalized promotions and remarketing campaigns to increase purchase frequency.

Task 4: Recommendation & Review Insights

Objective

Evaluate whether customer reviews and personalized recommendations significantly influence shopping satisfaction.

Key Findings

- Statistical analysis found **no significant relationship** between **Recommendation Helpfulness** and **Shopping Satisfaction** ($\chi^2 = 2.893$, $p = 0.9409$).
- **Review Reliability** also demonstrated an extremely weak relationship with Shopping Satisfaction (**Spearman's $\rho = -0.028$, $p = 0.4369$**).
- Customer perceptions of recommendation helpfulness were relatively balanced across **Yes**, **Sometimes**, and **No**, indicating mixed user experiences rather than a consistently positive or negative perception.
- Overall, recommendation systems and review reliability did not emerge as primary drivers of customer satisfaction within the surveyed population.

Business Impact

- Investments focused solely on improving recommendation algorithms are unlikely to substantially increase customer satisfaction.
- Core customer experience factors—including product quality, pricing, delivery performance, and customer service—are likely to provide greater business value.
- Recommendation systems should continue evolving as supporting features rather than primary customer satisfaction initiatives.

Task 5: Visualization & Reporting

Objective

Communicate analytical findings through business-oriented visualizations that support executive decision-making.

Key Findings

- Product category analysis highlighted **Clothing and Fashion (450 purchases)** as the strongest-performing category, while purchase activity remained well distributed across other major product groups.
- Browsing behavior analysis showed a balanced customer engagement profile, with **Rarely (231)** representing the largest browsing group, followed by **Few Times A Month (200)**, **Few Times A Week (186)**, and **Multiple Times A Day (183)**.
- Shopping satisfaction ratings remained evenly distributed across all five rating levels, indicating varied customer experiences rather than a strongly satisfied or dissatisfied customer base.
- Heatmap visualization confirmed the absence of a strong relationship between recommendation helpfulness and customer satisfaction, reinforcing the statistical findings obtained in Task 4.

Business Impact

- Business dashboards enable rapid identification of customer behavior patterns and support data-driven decision-making.
- Executive visualizations simplify complex customer information into actionable insights that can be shared across business teams.
- Interactive reporting improves communication between technical analysts and business stakeholders.

Overall Business Recommendations

Based on the combined findings from all analytical tasks, the following strategic recommendations are proposed.

1. Strengthen High-Demand Product Categories

- Continue prioritizing **Clothing and Fashion**, which recorded the highest purchase volume (**450 purchases**), through inventory optimization and targeted promotional campaigns.
- Maintain balanced inventory across **Home and Kitchen**, **Beauty and Personal Care**, and **Groceries and Gourmet Food**, as customer demand remains relatively evenly distributed.
- Reclassify products currently grouped under "**Others**" to obtain more granular customer purchasing insights.

2. Improve Customer Retention Through Segmentation

- Reward high-value customers through loyalty memberships, exclusive discounts, and personalized offers.
- Develop targeted recovery campaigns for At-Risk Customers to improve satisfaction before customer churn occurs.
- Encourage Satisfied Occasional Buyers to purchase more frequently through personalized recommendations and limited-time offers.

3. Improve the Overall Customer Experience

The statistical analysis indicates that recommendation systems alone are unlikely to significantly improve customer satisfaction.

Priority should instead be given to improving:

- Product quality
- Delivery efficiency
- Customer support responsiveness
- Competitive pricing
- Order fulfillment experience

These areas are expected to have a greater long-term impact on customer satisfaction and retention.

4. Refine the Recommendation Engine

Although recommendation helpfulness was not significantly associated with shopping satisfaction, recommendation systems remain valuable for customer engagement.

Future improvements should include:

- Better personalization using browsing and purchase history.
- Continuous evaluation of recommendation relevance.
- Integration of customer feedback into recommendation models.

5. Leverage Business Intelligence for Continuous Improvement

- Continuously monitor customer behavior using interactive dashboards.
- Track key performance indicators such as customer satisfaction, purchase frequency, and retention rates.

- Use data-driven insights to support strategic planning and continuous improvement initiatives across Amazon's customer experience ecosystem.

Conclusion

This project demonstrates how end-to-end data analytics can convert raw customer survey data into meaningful business intelligence. Through comprehensive data preparation, exploratory analysis, customer segmentation, statistical validation, and visualization, the analysis identified measurable opportunities to strengthen customer retention, optimize product strategy, and improve the overall shopping experience. While personalized recommendations and review reliability showed limited influence on customer satisfaction, the findings emphasize that sustained improvements in product quality, service efficiency, and customer engagement are more likely to drive long-term customer satisfaction and business growth.